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# Evaluation of an Ultrasonic Sensing Wake-Up Unit for Event-Driven Water Monitoring Systems

Sarah Didehvary  
Linn Hamadé

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Faculty of Natural Science

**Author**

Sarah Didehvary  
Linn Hamadé

**Title**

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**Supervisor**

Fredrik Frisk

**Examiner**

Åke Arvidsson

**Abstract**

Water-level monitoring systems often rely on continuously active sensing and imaging components, resulting in unnecessary energy consumption when relevant events occur infrequently. This thesis investigates a low-power wake-up approach for water-level monitoring based on ultrasonic sensing and event-driven system activation.

An embedded wake-up unit was evaluated under controlled laboratory conditions. The prototype used for the evaluation was based on an ESP32-S3 microcontroller and an ultrasonic distance sensor. The system operated by periodically waking from deep sleep, performing distance measurements, and activating an external camera subsystem only when predefined threshold conditions were detected.

The evaluation focused on power consumption, wake-up behavior, measurement stability, and event-detection performance under different test conditions. The results showed that the proposed approach reduced unnecessary subsystem activity while maintaining monitoring functionality. Ultrasonic sensing provided stable measurements under still-water, disturbed-water, and nearby reflective-surface conditions. Reliability decreased primarily during partial-obstruction tests, where incorrect reflections resulted in unstable measurements and inconsistent event detection.

The findings demonstrate that event-driven operation combined with ultrasonic sensing can provide a practical and energy-efficient solution for water-level monitoring applications. The evaluated architecture demonstrated reliable detection under most tested conditions with relatively low hardware complexity and processing requirements, suggesting its suitability for low-power monitoring applications.

**Keywords**

Low-power systems, Event-driven sensing, Ultrasonic sensor, Embedded systems, Water monitoring

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# 1. Introduction

Water-level monitoring systems used in tanks, pipes, and drainage infrastructure often need to operate for extended periods with limited maintenance. Changes such as rising water levels, buildup formation, or partial blockages are not always easily visible and may remain undetected until maintenance becomes necessary or operational failures occur.

Many existing monitoring solutions rely on manual inspection or continuously active sensing systems. Although such systems provide detailed monitoring and high responsiveness, they consume considerable amounts of energy and are therefore less suitable for battery-powered or remote applications. Camera-based systems face similar challenges because image capture, processing, and storage require significant power even when relevant events occur infrequently.

Research has shown that event-driven operation and selective subsystem activation can reduce unnecessary energy consumption in embedded monitoring systems [1], [2]. In these systems, higher-power subsystems remain inactive until a predefined event condition is detected.

This thesis investigates an ultrasonic-based wake-up unit developed for an existing camera-based monitoring system at Walthero. The evaluated prototype consists of an ESP32-S3 microcontroller and an ultrasonic distance sensor that periodically wake from deep sleep to evaluate water-level conditions and activate image capture only when required.

## 1.1. Problem Background and Motivation

The existing monitoring system developed by Walthero remains continuously active to ensure event capture, although relevant events occur relatively infrequently. As a result, energy is consumed even during long periods without activity, limiting the suitability of battery-powered operation and increasing maintenance requirements.

The motivation for this work was therefore to investigate whether ultrasonic sensing combined with periodic wake-up operation could provide reliable event detection while minimizing unnecessary energy consumption.

## 1.2. Scope and Objectives

The objective of this thesis was to experimentally evaluate an ultrasonic-based wake-up unit for low-power water-level monitoring under controlled laboratory conditions. The study focused on power consumption, wake-up behavior, measurement stability, and event detection. Advanced data analytics, cloud-based processing, machine-learning techniques, and long-term field deployment were outside the scope of this work.

## 1.3. Research Questions

Based on the objectives of the thesis, the following research questions were defined:

- **RQ1:** How reliably can an ultrasonic-based wake-up unit detect water-level-related events and trigger system activation under realistic monitoring conditions?
- **RQ2:** How does ultra-low-power operation affect detection accuracy, responsiveness, and overall system performance?

## 1.4. Research Methodology

The research consisted of a literature review and an experimental evaluation of an existing prototype developed within the associated Systems Engineering project. The original Systems Engineering project investigated wake-up concepts for low-power monitoring systems and first considered radar-based sensing as a possible approach. During prototype development, ultrasonic sensing was selected for implementation and experimental evaluation because of its lower complexity, lower cost and suitability for the available test environment. This thesis evaluates the ultrasonic-sensor prototype that emerged from the broader Systems Engineering project.

## 1.5. Thesis Structure

The remainder of the thesis is organized as follows. Chapter 2 presents the literature review and analysis of previous work. Chapter 3 describes the system design and implementation. Chapter 4 presents the experimental setup and testing methodology. Chapter 5 presents the results. Chapter 6 discusses the findings, limitations, and possible improvements. Finally, Chapter 7 presents the conclusions of the thesis.

## 2. Literature Review

### 2.1. Literature Search Methodology

Table 1 summarizes the literature search process used to identify research related to low-power embedded systems, wake-up mechanisms, ultrasonic sensing, and environmental monitoring. The search was conducted using the HKR SuperSearch platform (EBSCOhost), which provides access to databases including IEEE Xplore and ScienceDirect.

The search process began with broad keywords that were gradually refined into more specific search combinations. ChatGPT (OpenAI) was used to assist in generating keywords based on the project scope and system requirements.

Articles were initially screened using titles and abstracts. Selection was then narrowed based on publication year (2022 or later), peer-reviewed status, practical relevance, and the presence of experimental or implementation-based results. Greater emphasis was placed on studies involving prototype development, hardware integration, and experimental evaluation rather than purely theoretical or simulation-based work.

For refined searches returning more than 100 results, only the first 100 articles sorted by relevance were reviewed. Since the system concept and hardware platform had already been established within the associated project, the literature review focused on identifying previous work relevant to the chosen approach. When multiple relevant papers were found for a particular keyword search, the study that best matched the functionality, sensing method, application domain, and technical characteristics of the evaluated system was selected for detailed discussion. A total of seven studies were included in the review. The review was conducted as a focused literature review intended to support the selected technical approach rather than as a comprehensive survey of all available research within the field.



**Table 1.** Literature search and keyword refinement process

| Category                 | Broad Key-word                    | Hits  | Refined Key-word                                  | Hits | Selected Study | DB            |
|--------------------------|-----------------------------------|-------|---------------------------------------------------|------|----------------|---------------|
| Low-Power Systems        | ESP32 deep sleep                  | 26    | ESP32 deep sleep environmental monitoring         | 6    | [1]            | IEEE          |
| Camera Optimization      | camera low power monitoring       | 1654  | sensor triggered camera activation                | 128  | [2]            | IEEE          |
| Wake-Up Systems          | wake-up sensor network            | 2147  | event driven sensor network low power             | 497  | [3]            | IEEE          |
| Embedded Monitoring      | embedded monitoring system        | 40279 | embedded environment monitoring system            | 7909 | [4]            | IEEE          |
| Ultrasonic Monitoring    | ultrasonic water level monitoring | 1074  | ultrasonic water level monitoring embedded system | 40   | [5]            | IEEE          |
| Environmental Monitoring | smart water tank monitoring       | 63    | water tank monitoring embedded system             | 18   | [6]            | ScienceDirect |
| Radar Sensing            | radar level measurement           | 18958 | FMCW radar liquid level measurement               | 22   | [7]            | IEEE          |

## 2.2. Overview of Selected Studies

The selected studies address several themes relevant to this work, including low-power embedded monitoring, wake-up mechanisms, event-driven operation, and non-contact sensing. The following sections summarize the main findings and approaches reported in the literature.

## 2.3. Analysis of Existing Approaches

### 2.3.1. Low-Power and Event-Driven Monitoring

A recurring approach in low-power monitoring systems is event-driven operation, where parts of the system remain inactive until a relevant environmental change is detected.

Ghosh et al. [1] demonstrated that deep-sleep modes, event-driven wake-ups, and lightweight wireless communication can significantly extend the operational lifetime of ESP32-based environmental monitoring systems. Chen et al. [3] discussed wake-up receiver architectures that reduce standby power consumption while maintaining fast responses to external events.

Ma et al. [2] demonstrated selective subsystem activation in a fire-monitoring system, where cameras remained inactive until abnormal conditions were detected by low-power smoke sensors. Similarly, Shabanski and Marinov [4] presented a battery-powered environmental monitoring system using threshold-based wake-up signaling to activate more power-intensive measurements only when changes in environmental parameters were detected.

Together, these studies demonstrate that deep-sleep operation, wake-up mechanisms, and selective activation are effective approaches for reducing energy consumption in embedded monitoring systems.

### 2.3.2. Non-Contact Sensing Approaches

Non-contact sensing is widely used in environmental monitoring systems where direct sensor exposure can increase maintenance requirements or reduce sensor reliability.

Sasikala et al. [5] developed an IoT-based reservoir monitoring system using ultrasonic sensing and ESP32-based control for real-time monitoring. Khan et al. [6] similarly implemented an ultrasonic monitoring system capable of water-level measurement, leakage detection, and real-time monitoring through embedded sensor integration.

As a comparison, Li et al. [7] proposed an FMCW radar liquid-level measurement system using FFT and CZT signal-processing techniques to improve measurement accuracy. Although radar-based approaches generally achieve higher accuracy, they require more advanced signal processing, greater computational resources, and more complex hardware integration.

This trade-off between measurement performance and system complexity makes ultrasonic sensing attractive for low-power embedded monitoring applications, where low computational requirements and simple hardware integration are important design considerations.

Another observation from the reviewed studies is that each study primarily addresses a specific subsystem or application, such as wake-up mechanisms, low-power sensing, camera activation, or level measurement. Consequently, relatively few of the reviewed works evaluate complete monitoring architectures under varying environmental conditions, making it difficult to assess their suitability for long-term deployment.

## 2.4. Research Gap and Contribution

The reviewed studies show significant progress in low-power monitoring systems, wake-up architectures, and non-contact sensing technologies. However, relatively few studies combine ultrasonic sensing, event-driven operation, and selective subsystem activation within a single experimentally evaluated low-power monitoring system. In addition, many existing studies focus on system design or application functionality without investigating the effects of periodic wake-up operation on power consumption, responsiveness, and event-detection reliability.

This thesis addresses these limitations through the experimental evaluation of an ultrasonic-based wake-up unit operating within a low-power event-driven monitoring architecture. Rather than focusing on full system development, the main contribution is the assessment of measurement stability, event-detection behavior, wake-up timing, and power consumption under different operating conditions. The results provide insight into the suitability of ultrasonic

sensing and periodic wake-up operation for low-power water-level monitoring applications.

## 3. Method

This chapter describes the evaluated monitoring system and the methodology used for its experimental assessment. The work focused on a lightweight embedded wake-up unit for water-level monitoring based on an ESP32-S3 microcontroller and an ultrasonic distance sensor.

The wake-up unit was designed to minimize energy consumption by remaining in deep sleep between measurement intervals and activating additional functionality only when predefined conditions were detected.

### 3.1. Research Methodology

The research methodology consisted of four stages:

- I. Literature review and analysis of existing research
- II. Experimental testing of the prototype under controlled conditions
- III. Collection of measurement data related to power consumption, wake-up behavior, and detection performance
- IV. Analysis and interpretation of experimental results and evaluation of the research questions

The hardware and software development of the prototype were carried out within the associated Systems Engineering course project. This thesis focused on the experimental assessment of the completed wake-up unit and the analysis of its performance with respect to the research questions.

The literature review was used to identify relevant previous work and support the interpretation of the experimental results. The completed system was then evaluated through controlled experiments focusing on power consumption, timing behavior, measurement stability, and detection performance.

### 3.2. System Overview

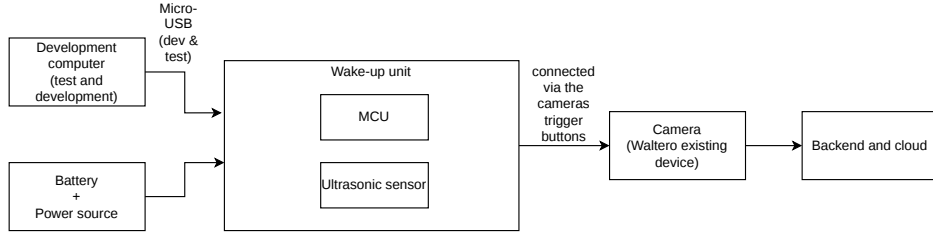
Figure 1 illustrates the architecture of the monitoring system evaluated in this thesis. The system consists of a battery-powered wake-up unit containing an ESP32-S3 microcontroller and an ultrasonic sensor, together with an existing camera device and cloud-based backend infrastructure.

The ultrasonic sensor measures the distance to the monitored water surface and provides the measurement data to the ESP32-S3. The microcontroller periodically wakes from deep sleep, performs a measurement cycle, and evaluates the measured distance against a predefined threshold value.

If no event condition is detected, the system returns to deep sleep to minimise power con-

sumption. When the threshold condition is met, the ESP32-S3 activates the trigger connection linked to the camera device. The camera then captures images and transfers the collected data to the backend and cloud infrastructure for further processing and storage.

During development of the prototype within the associated Systems Engineering project, the wake-up unit was connected to a development computer through a Micro-USB interface for programming and debugging. The same interface was used during the experimental evaluation performed in this thesis.



**Figure 1.** System architecture of the proposed event-driven wake-up monitoring system.

### 3.3. Microcontroller Configuration

The evaluated prototype was based on an ESP32-S3 DevKitC-1 microcontroller configured for periodic wake-up operation and low-power monitoring.

The ESP32-S3 uses a timer-based wake-up mechanism to periodically wake from deep sleep and perform a measurement cycle. In the implemented prototype, the wake-up interval was set to 60 minutes. During each cycle, the ultrasonic sensor performs a distance measurement which is compared against a predefined threshold value. If the measured distance indicates an event condition, the ESP32-S3 activates the required subsystems for image capture and event handling. Otherwise, the system returns directly to deep sleep until the next scheduled wake-up interval.

The threshold-based triggering mechanism was implemented using a predefined distance limit of 50 cm. Measurements below this threshold were treated as event conditions requiring subsystem activation. The permissible threshold range was defined by Waltero based on the intended monitoring application. Several candidate threshold values within this range were evaluated during system testing, and a threshold of 50 cm was selected because it provided the most consistent event-detection behavior under the tested conditions.

Periodic wake-up operation was considered sufficient for the intended monitoring application because water-level changes generally occur slowly relative to the wake-up and processing time of the microcontroller.

### 3.4. Ultrasonic Sensor and Measurement Principle

The AJ-SR04M ultrasonic sensor was used for non-contact distance measurement. The sensor operates using acoustic time-of-flight, where an ultrasonic pulse is transmitted and the reflected signal delay is converted into a distance value.

Ultrasonic sensing was used in the evaluated prototype because it provides a relatively simple and lightweight solution for short-range environmental monitoring applications while requiring limited processing and hardware complexity.

Environmental factors including surface movement, nearby objects, reflections, and partial obstructions can influence the stability of the measurement. To evaluate these effects, measurements were performed under different environmental conditions during the experimental phase.

### 3.5. Camera Activation Strategy

The camera subsystem remains disabled during normal operation and is only activated when the measured distance drops below the predefined threshold value.

When an event condition is detected, the ESP32-S3 enables the camera subsystem for image capture and event handling. After the capture process is completed, the subsystem is disabled and the system returns to deep sleep.

This approach limits unnecessary subsystem activity while still allowing visual documentation of detected events.

### 3.6. Power Consumption Evaluation Method

Power consumption was evaluated experimentally by measuring the current consumption of the system during its three primary operating states: deep sleep, ultrasonic measurement, and processing.

The average current consumption was calculated using

$$I_{\text{avg}} = \frac{I_{\text{sleep}} \cdot t_{\text{sleep}} + I_{\text{measure}} \cdot t_{\text{measure}} + I_{\text{process}} \cdot t_{\text{process}}}{T}$$

where

- $I_{\text{avg}}$  is the average current consumption of the system
- $I_{\text{sleep}}$  is the current consumption during deep-sleep operation
- $I_{\text{measure}}$  is the current consumption during ultrasonic measurement
- $I_{\text{process}}$  is the current consumption during processing and threshold evaluation

- $t_{\text{sleep}}$ ,  $t_{\text{measure}}$ , and  $t_{\text{process}}$  are the corresponding operating durations
- $T$  is the total operating cycle duration

The total operating cycle duration was calculated as

$$T = t_{\text{sleep}} + t_{\text{measure}} + t_{\text{process}}$$

The average power consumption was calculated using

$$P_{\text{avg}} = V_{\text{supply}} \cdot I_{\text{avg}}$$

where

- $P_{\text{avg}}$  is the average power consumption of the system
- $V_{\text{supply}}$  is the supply voltage
- $I_{\text{avg}}$  is the average current consumption of the system

The calculations were used to support analysis of the measured power behavior of the evaluated prototype under the tested operating conditions. They are intended to provide an understanding of the contribution of different operating states rather than to predict long-term battery lifetime in real deployment scenarios.

## 4. Experimental Setup

### 4.1. Hardware Platform

The experimental setup was built using the ESP32-S3 DevKitC-1 development board together with the AJ-SR04M ultrasonic sensor for non-contact distance measurement.

The hardware platform was used to test ultrasonic distance measurement, wake-up behavior, threshold-based event detection, and subsystem activation during low-power operation.

### 4.2. Measurement Methodology

Current consumption was measured separately for each operating state to evaluate how different subsystems contributed to the total power usage. Measurements were performed during deep-sleep operation, ultrasonic measurement, and processing.

Current measurements were obtained using an INA226 current and power monitoring IC from Texas Instruments connected in series between the battery supply and the ESP32-S3. The measured current values were used to determine the current consumption of each operating

state.

Power consumption was calculated using

$$P = V \cdot I$$

where

- $P$  is the power consumption
- $V$  is the supply voltage
- $I$  is the measured current consumption

### 4.3. Wake-Up Timing Evaluation

The wake-up behavior of the ESP32-S3 was evaluated by measuring the time required to wake from deep sleep, complete a measurement cycle, and respond to a threshold condition. Timing measurements were obtained by repeatedly activating the system under identical conditions and observing the elapsed time for each stage of operation. The measurements were repeated five times to assess the consistency of the periodic wake-up behavior.

### 4.4. Detection Test Design

Table 2 presents the experimental test conditions used during system evaluation.

Measurements were performed using water contained in a plastic container positioned below the ultrasonic sensor. Testing was carried out at distances of 0.5 m, 1.5 m, and 3.0 m under different environmental conditions to evaluate measurement stability and detection behavior.

The experiments included measurements using still water, disturbed water with manually created ripples, nearby reflective surfaces, and partial obstruction of the sensing path. These conditions were selected to investigate how environmental disturbances influenced ultrasonic measurements and threshold-triggered subsystem activation.

Each distance and environmental condition presented in Table 2 was tested five times to evaluate measurement consistency, event-detection behavior, and the occurrence of false activations under repeated operating conditions.



**Table 2.** Experimental test matrix for detection evaluation

| <b>Distance (m)</b> | <b>Condition</b>          | <b>Test Description</b>                         |
|---------------------|---------------------------|-------------------------------------------------|
| 0.5                 | Still water               | Flat and undisturbed water surface              |
| 0.5                 | Disturbed water           | Water surface with manually created ripples     |
| 0.5                 | Nearby reflective surface | Metal plate placed beside plastic container     |
| 0.5                 | Partial obstruction       | Cardboard sheet partially blocking sensing path |
| 1.5                 | Still water               | Flat and undisturbed water surface              |
| 1.5                 | Disturbed water           | Water surface with manually created ripples     |
| 1.5                 | Nearby reflective surface | Metal plate placed beside plastic container     |
| 1.5                 | Partial obstruction       | Cardboard sheet partially blocking sensing path |
| 3.0                 | Still water               | Flat and undisturbed water surface              |
| 3.0                 | Disturbed water           | Water surface with manually created ripples     |
| 3.0                 | Nearby reflective surface | Metal plate placed beside plastic container     |
| 3.0                 | Partial obstruction       | Cardboard sheet partially blocking sensing path |

## 5. Result

The results include power consumption, wake-up behavior, ultrasonic measurement stability, detection performance, and evaluation of the research questions.

### 5.1. Power Consumption Results

Table 3 presents the measured current consumption and calculated power consumption for the main operating states of the system. The measurements were used to evaluate how each operating state contributed to the total power consumption during system operation.

The lowest power consumption was measured during deep-sleep operation, where the system consumed only 0.018 mA. During ultrasonic measurement, the current increased to 12.53 mA, while the processing state produced the highest measured current consumption at 34.06 mA.

The external camera system was not included in the experimental measurements. The evaluation instead focused on the wake-up unit, threshold detection behavior, periodic wake-up operation, and low-power system performance.

**Table 3.** Measured current consumption and calculated power consumption for each operating state

| Operating State        | Active Components                    | Battery Voltage (V) | Current (mA) | Power (mW) |
|------------------------|--------------------------------------|---------------------|--------------|------------|
| Deep Sleep             | RTC timer only                       | 6                   | 0.018        | 0.108      |
| Ultrasonic Measurement | ESP32-S3, ultrasonic sensor          | 6                   | 12.53        | 75.19      |
| Processing State       | ESP32-S3, threshold evaluation logic | 6                   | 34.06        | 204.3      |

### 5.2. Wake-Up Behavior

Table 4 summarizes the measured wake-up behavior of the ESP32-S3 during periodic operation. The measurements were used to evaluate how quickly and consistently the system responded during repeated wake-up cycles.

The ESP32-S3 required approximately 500 ms to wake from deep sleep. The complete measurement cycle required approximately 1500 ms, while the total threshold detection response time was approximately 2050 ms.

The results showed stable timing behavior during repeated tests, indicating that the periodic wake-up operation provided predictable system responsiveness while still allowing the ESP32-S3 to remain in deep sleep during inactive periods.

**Table 4.** Wake-up and response measurements

| Measurement                       | Measured Value (ms) | Observation                      |
|-----------------------------------|---------------------|----------------------------------|
| Wake-up time from deep sleep      | 500                 | Stable across repeated tests     |
| Measurement cycle duration        | 1500                | Stable across repeated tests     |
| Threshold detection response time | 2050                | Consistent across repeated tests |

### 5.3. Ultrasonic Measurement Stability

Table 5 presents the measured distances obtained during repeated tests under different environmental conditions. Each test condition was repeated five times. For conditions where measurement variations occurred, the reported values represent the range observed across the repeated measurements.

The results show that the measured distances remained relatively stable under still-water, disturbed-water, and nearby reflective-surface conditions. Here, stability refers to the consistency of repeated measurements rather than their absolute accuracy, especially at longer sensing distances.

The largest deviations occurred during the partial-obstruction tests, where the reflected ultrasonic signal was partially blocked. This resulted in unstable and incorrect distance measurements, indicating reduced measurement reliability when a clear sensing path was not maintained.

**Table 5.** Ultrasonic measurement stability results

| Distance (m) | Condition                 | Measured Distance (m) | Observation          |
|--------------|---------------------------|-----------------------|----------------------|
| 0.5          | Still water               | 0.53                  | Stable measurement   |
| 0.5          | Disturbed water           | 0.5-0.51              | Stable measurement   |
| 0.5          | Nearby reflective surface | 0.58                  | Stable measurement   |
| 0.5          | Partial obstruction       | 0.3                   | Unstable measurement |
| 1.5          | Still water               | 1.52                  | Stable measurement   |
| 1.5          | Disturbed water           | 1.53                  | Stable measurement   |
| 1.5          | Nearby reflective surface | 1.57                  | Stable measurement   |
| 1.5          | Partial obstruction       | 0.3                   | Unstable measurement |
| 3.0          | Still water               | 3.2                   | Stable measurement   |
| 3.0          | Disturbed water           | 2.8-3.0               | Stable measurement   |
| 3.0          | Nearby reflective surface | 2.9-3.1               | Stable measurement   |
| 3.0          | Partial obstruction       | 0.3                   | Unstable measurement |

### 5.4. Detection Performance

The results presented in Table 6 were used to evaluate how consistently the system responded when the measured distance dropped below the selected threshold value. In this context, “Event Detected” indicates that the system correctly identified the threshold condition and triggered the intended response, while “False Trigger” refers to unintended activation caused by incorrect

distance measurements.

The system successfully detected events during still-water, disturbed-water, and nearby reflective-surface tests without producing false triggers. In contrast, the partial-obstruction tests resulted in missed detections and unintended activations due to incorrect ultrasonic measurements.

**Table 6.** Threshold detection and event response results

| Test Condition            | Event Detected | False Trigger | Observation                          |
|---------------------------|----------------|---------------|--------------------------------------|
| Still water               | Yes            | No            | Stable readings                      |
| Disturbed water           | Yes            | No            | $\pm 0.01$ m measurement variation   |
| Nearby reflective surface | Yes            | No            | $\pm 0.02$ m measurement variation   |
| Partial obstruction       | No             | Yes           | Obstruction blocked reflected signal |

## 5.5. Research Question Evaluation

The results presented in this chapter are used to evaluate the research questions defined in Section 1.3.

**RQ1: How reliably can an ultrasonic-based wake-up unit detect water-level-related events and trigger system activation under realistic monitoring conditions?**

RQ1 is evaluated using the ultrasonic measurement stability results and the threshold detection tests presented in Tables 5 and 6. Particular attention is given to measurement stability, detection consistency, and false activations under different environmental conditions.

**RQ2: How does ultra-low-power operation affect detection accuracy, responsiveness, and overall system performance?**

RQ2 is evaluated using the power-consumption measurements and wake-up timing measurements presented in Tables 3 and 4. The evaluation focuses on how deep-sleep operation and periodic wake-up behavior affect system responsiveness and overall performance while maintaining low energy consumption.

## 6. Discussion

This chapter discusses the experimental results and examines how the design choices influenced system behavior and performance. The findings are related to the research questions, and the limitations and potential improvements of the proposed monitoring approach are considered.

### 6.1. Event-Driven and Periodic Wake-Up Operation

The implemented wake-up strategy was effective in reducing energy consumption by limiting system activity to short measurement intervals. Since the ESP32-S3 remained in deep sleep for most of the operating period, power-intensive processing occurred only when required.

The large difference between deep-sleep and active-state current consumption shows the importance of minimizing active operating time in low-power embedded systems. This observation is consistent with previous studies that identified deep-sleep operation and event-driven activation as effective approaches for reducing energy consumption [1], [3].

The measured wake-up and processing times contributed only a small portion of the total response time. Since the system performed measurements once every 60 minutes, the selected wake-up interval was the primary factor affecting responsiveness rather than the wake-up and processing times themselves. Consequently, the reduction in power consumption achieved through extended deep-sleep operation came at the cost of a potentially longer delay before an event could be detected. This illustrates the trade-off between energy efficiency and responsiveness that is common in low-power monitoring systems.

The effectiveness of the approach depends on reliable event detection. Frequent false activations would increase active operating time and reduce the expected energy savings.

### 6.2. Ultrasonic Sensing for Water-Level Monitoring

Ultrasonic sensing proved suitable for short-range water-level monitoring due to its simple implementation and low processing requirements. The results indicate that stable measurements can be achieved under controlled conditions, particularly at shorter sensing distances.

The observed measurement variations under disturbed-water, reflective-surface, and obstruction conditions demonstrate the sensitivity of ultrasonic sensing to environmental influences. These effects became more noticeable as sensing distance increased, suggesting that measurement reliability depends on both environmental conditions and signal quality.

The most significant limitation was observed during partial obstruction testing. In these scenarios, unreliable reflections produced incorrect distance measurements and reduced detection reliability. This highlights the importance of maintaining a clear sensing path when ultrasonic sensors are used for event-based monitoring applications.

Although radar-based systems may provide improved robustness under some conditions, the results suggest that ultrasonic sensing remains a practical alternative when simplicity, low hardware complexity, and low processing requirements are prioritized.

### 6.3. Threshold-Based Detection

The threshold-based detection approach provided a simple and computationally efficient method for event identification. Because only basic distance comparisons were required, processing overhead remained low and contributed minimally to system complexity.

The results indicate that fixed-threshold detection was sufficient under stable operating conditions. Even so, the obstruction tests demonstrated that incorrect measurements can lead to unreliable triggering behavior when the reflected ultrasonic signal becomes unstable.

This represents a common trade-off in low-power embedded systems. Simpler detection methods reduce computational requirements and energy consumption but generally provide lower robustness under challenging environmental conditions. More advanced filtering or adaptive threshold techniques could improve reliability without significantly increasing processing requirements.

### 6.4. Research Question Discussion

**RQ1 examined how reliably the ultrasonic-based wake-up unit could detect water-level-related events under realistic monitoring conditions.**

In this work, reliability refers to measurement stability, detection consistency, and the occurrence of incorrect activations.

The results indicate that reliable event detection was achieved under most tested conditions. Partial obstructions reduced detection consistency, demonstrating that ultrasonic sensing performance depends on maintaining a clear sensing path.

**RQ2 examined how low-power operation affected responsiveness, detection behavior, and system performance.**

The results indicate that deep-sleep operation enabled substantial energy savings without noticeably affecting detection behavior under normal operating conditions. However, periodic wake-up operation introduced a trade-off between energy efficiency and responsiveness, since events could only be detected during scheduled measurement cycles.

### 6.5. System Limitations

The prototype was evaluated under controlled laboratory conditions and therefore does not fully represent long-term deployment scenarios. Environmental factors such as temperature

variation, humidity, contamination, and long-term sensor aging were not investigated.

The system also relied on fixed threshold values and simple local processing. While these design choices reduced complexity and power consumption, they may limit robustness under changing environmental conditions.

The obstruction experiments demonstrated that ultrasonic sensing performance depends on reliable signal reflections.

## **6.6. Future Improvements**

Future work could focus on improving system robustness under varying environmental conditions through adaptive threshold adjustment, signal filtering, and additional sensing modalities. Such improvements may help reduce the impact of obstructions, strong reflections, and other sources of measurement uncertainty.

Long-term field testing would provide valuable insight into system reliability and environmental influences outside controlled laboratory conditions. Wireless communication could further extend functionality by enabling remote monitoring and periodic status reporting. Such functionality could simplify the detection of subsystem failures, unexpected shutdowns, and power loss in long-term autonomous deployments.

## 7. Conclusion

This thesis investigated the use of ultrasonic sensing and periodic wake-up operation for low-power water-level monitoring.

The results showed that the ultrasonic-based wake-up unit was able to detect events related to water-level under still-water, disturbed-water, and reflective-surface conditions. Measurement stability and detection performance remained acceptable under most test conditions. Although partial-obstruction experiments demonstrated that ultrasonic sensing performance strongly depends on the quality of the reflected signal, blocked or unstable reflections resulted in incorrect distance measurements and unreliable event detection.

The study also showed that periodic wake-up operation reduced power consumption while maintaining sufficient responsiveness for applications involving gradual water-level changes. The measured wake-up times indicated that low-power operation introduced only a limited processing delay, although the selected measurement interval represents a trade-off between responsiveness and energy efficiency.

The main contribution of this work is the experimental evaluation of measurement stability, event-detection behavior, wake-up timing, and power consumption for an ultrasonic-based wake-up unit operating within a lightweight event-driven monitoring architecture. The findings indicate that water-level-related events can be detected with relatively low hardware complexity and processing requirements while reducing unnecessary energy consumption.

Future work could focus on improving system robustness through adaptive threshold adjustment, signal filtering, and additional sensing methods. Long-term field testing under real operating conditions would provide further insight into system reliability and environmental influences outside controlled laboratory conditions.



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